

Distributed Optimization over Undirected Graphs with Nonlinear Edge-Based Coupling

1 Preliminaries and Notation

Throughout this section, we introduce the notation and fundamental mathematical concepts from graph theory and real analysis used in this study.

1.1 Notation

Let $\mathbb{R}$, $\mathbb{R}^n$, and $\mathbb{R}^{n \times m}$ denote the sets of real numbers, real n -dimensional vectors, and real $n \times m$ matrices, respectively. The sets $\mathbb{R}_{>0}$, $\mathbb{R}_{\geq 0}$, and $\mathbb{Z}_{>0}$ represent the positive real, nonnegative real, and positive integer numbers, respectively. The transpose of a vector or matrix is denoted by $(\cdot)^T$. The identity matrix of dimension n is denoted by I_n . The column vectors with all entries equal to one and zero in $\mathbb{R}^N$ are denoted by $\mathbf{1}_N$ and $\mathbf{0}_N$, respectively. The symbol $\|\cdot\|$ denotes the Euclidean norm for vectors and the Frobenius norm for matrices. The i th eigenvalue of a matrix A is denoted by $\lambda_i(A)$, and $\otimes$ denotes the Kronecker product.

For vectors $u_1, \dots, u_m$, we write $u = (u_1, \dots, u_m)$ to denote the aggregated vector. In a networked system, local variables of agent i are denoted using a superscript; for example, x^i denotes the state of agent i . If $p^i \in \mathbb{R}^d$ is a variable of agent i , the aggregated variable of the N -agent network is written as $p = (p^1, \dots, p^N) \in (\mathbb{R}^d)^N$. We define $\text{diag}(\mathbb{R}^N) = \{x \in \mathbb{R}^N \mid x_1 = \dots = x_N\} \subset \mathbb{R}^N$ as the agreement subspace in $\mathbb{R}^N$. For a finite set S , its cardinality is denoted $|S|$.

For a symmetric matrix P , $P > 0$ ($P < 0$) denotes that P is positive (negative) definite. The absolute value of a constant $\alpha \in \mathbb{R}$ is denoted $|\alpha|$. For a column vector $\xi = [\xi_1, \xi_2, \dots, \xi_p]^T$, its q -norm for $q > 0$ is defined as

$$\|\xi\|_q = \left(\sum_{i=1}^p |\xi_i|^q \right)^{1/q},$$

and $\|A\|_\infty$ denotes the infinity-norm of a matrix A . For $y \in \mathbb{R}$, we define the signum function and its corresponding set-valued map as follows:

$$\text{sign}(y) := \begin{cases} 1, & y > 0, \\ 0, & y = 0, \\ -1, & y < 0. \end{cases} \quad \mathbf{sgn}(y) := \begin{cases} \{\text{sign}(y)\}, & y \neq 0, \\ [-1, 1], & y = 0, \end{cases}$$

The function $\mathbf{sgn}(y)$ is used to obtain the set-valued map for a discontinuous system. In case of a vector argument, $y = [y_1, y_2, \dots, y_m]^T \in \mathbb{R}^m$, these functions are defined as

$$\begin{aligned} \text{sign}(y) &:= [\text{sign}(y_1), \text{sign}(y_2), \dots, \text{sign}(y_m)]^T, \\ \mathbf{sgn}(y) &:= [\mathbf{sgn}(y_1), \mathbf{sgn}(y_2), \dots, \mathbf{sgn}(y_m)]^T. \end{aligned}$$

For any $x, y \in \mathbb{R}^n$, Young's inequality states that, for any $\varepsilon \in \mathbb{R}_{>0}$,

$$x^T y \leq \frac{1}{2\varepsilon} x^T x + \frac{\varepsilon}{2} y^T y.$$

1.2 Graph Theory

An undirected graph is denoted $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, \dots, N\}$ is the node set and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set, with each edge $\{i, j\} \in \mathcal{E}$ regarded as an unordered pair (i.e., $\{i, j\} = \{j, i\}$).

To define the incidence matrix, an *arbitrary reference orientation* is assigned to each edge: for edge $e_k = \{i, j\}$, one endpoint is arbitrarily designated the head and the other the tail. This orientation is introduced purely as a bookkeeping device to construct the matrix below and does *not* imply that the underlying graph is directed or that information/flow is restricted between i and j ; the resulting Laplacian matrix defined below is independent of this choice. The incidence matrix $D \in \mathbb{R}^{N \times |\mathcal{E}|}$ of $\mathcal{G}$ is the matrix whose columns are indexed by $\mathcal{E}$, with entries

$$D_{ik} = \begin{cases} +1, & \text{if node } i \text{ is the (arbitrarily assigned) head of edge } e_k, \\ -1, & \text{if node } i \text{ is the (arbitrarily assigned) tail of edge } e_k, \\ 0, & \text{if node } i \text{ is not incident to edge } e_k. \end{cases}$$

The degree matrix $\Delta(\mathcal{G}) \in \mathbb{R}^{N \times N}$ of $\mathcal{G}$ is a diagonal matrix whose i th diagonal entry equals the degree of node i (the number of edges incident to i), with all off-diagonal entries equal to 0. The adjacency matrix $A(\mathcal{G}) \in \mathbb{R}^{N \times N}$ of $\mathcal{G}$ has zero diagonal entries, and its (i, j) th entry equals 1 if nodes i and j are adjacent (i.e., $\{i, j\} \in \mathcal{E}$), and 0 otherwise.

The Laplacian matrix $L = [l_{ij}] \in \mathbb{R}^{N \times N}$ of $\mathcal{G}$ is defined as

$$L := DD^T = \Delta(\mathcal{G}) - A(\mathcal{G}),$$

which is symmetric positive semidefinite, and despite D itself depending on the chosen edge orientation the product $L = DD^T$ is invariant under the choice of reference orientation.

An undirected path from node i_1 to node i_s is a sequence of undirected edges (i_k, i_{k+1}) , $k = 1, \dots, s-1$. An undirected graph $\mathcal{G}$ is said to be connected if there exists an undirected path between every pair of distinct nodes.

1.3 Real Analysis

Let $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, and their aggregate be defined as

$$f(x) = \sum_{i=1}^N f_i(x).$$

The gradient of f_i is denoted $\nabla f_i(x)$, and that of the global cost is denoted $\nabla f(x)$. A set $\Omega \subseteq \mathbb{R}^p$ is convex if, for all $x, y \in \Omega$ and $\lambda \in [0, 1]$, $\lambda x + (1 - \lambda)y \in \Omega$. A differentiable function f is m -strongly convex on a convex set Ω if there exists $m > 0$ such that

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq m \|x - y\|^2, \quad (1)$$

which equivalently means $f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{m}{2} \|y - x\|^2$, and implies f admits a unique minimizer. A function f is locally Lipschitz if, for every $x \in \mathbb{R}^p$, there exists a neighborhood W of x and a constant $M \geq 0$ such that

$$\|f(y) - f(z)\| \leq M \|y - z\|, \quad \forall y, z \in W.$$

1.4 Useful Lemmas

Assumption 1. *The undirected graph $\mathcal{G}$ of the networked agents considered in this paper is connected.*

Lemma 1. *If Assumption 1 holds, then 0 is a simple eigenvalue of the Laplacian matrix L , with $\mathbf{1}$ as a corresponding right eigenvector, and all nonzero eigenvalues are positive. Furthermore, the nonzero smallest eigenvalue $\lambda_2(L)$ of L is*

$$\lambda_2(L) = \min_{x \neq 0, \mathbf{1}^T x = 0} \frac{x^T L x}{x^T x}.$$

Define $M = I_N - (1/N)\mathbf{1}\mathbf{1}^T$. It follows from Lemma 3 that 0 is a simple eigenvalue of M with $\mathbf{1}$ as the corresponding right eigenvector, i.e., $M\mathbf{1} = 0$. Further, $M^2 = M$ and $M^T = M$.

Lemma 2. *Under Assumption 1, for matrices M and D defined above, and for any vector $\xi = [\xi_1^T, \xi_2^T, \dots, \xi_N^T]^T \in \mathbb{R}^{Nn}$ with $\xi_i \in \mathbb{R}^n$, one has*

$$\|(M \otimes I)\xi\|_1 \leq \frac{N-1}{2} \|(D^T \otimes I)\xi\|_1. \quad (2)$$

Proof. It follows from the definition of M that:

$$\begin{aligned} \|(M \otimes I)\xi\|_1 &\leq \frac{1}{N} \sum_{i=1}^N \sum_{j=1, j \neq i}^N \|\xi_i - \xi_j\|_1 \\ &\leq \frac{1}{N} \sum_{i=1}^N \max_i \left\{ \sum_{j=1, j \neq i}^N \|\xi_i - \xi_j\|_1 \right\} \\ &= \max_i \left\{ \sum_{j=1, j \neq i}^N \|\xi_i - \xi_j\|_1 \right\} \\ &\leq \frac{N-1}{2} \sum_{i=1}^N \sum_{j \in \mathcal{N}_i} \|\xi_i - \xi_j\|_1 \\ &= \frac{N-1}{2} \|(D^T \otimes I)\xi\|_1. \end{aligned}$$

This completes the proof. □

2 Problem Statement and Equivalent Formulations

Consider a set of N agents interacting over a communication graph. Each agent $i \in \{1, 2, \dots, N\}$ is associated with a local cost function defined at each node, $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, whose input is the corresponding state vector $x^i \in \mathbb{R}^n$ of the agent. Let z be the aggregate vector defined as $z = (x^1, x^2, \dots, x^N)$. The global cost function is defined as

$$f(z) = \sum_{i=1}^N f_i(z).$$

The objective is to design a continuous-time distributed algorithm such that each agent can solve the consensus optimization problem

$$\min_{z \in \mathbb{R}^n} f(z) \quad (3)$$

using only its own information and information received from its neighboring agents as binary input of $\{+1, -1\}$.

2.1 Assumptions

Assumption 2. *The global cost function f is differentiable with Lipschitz gradients and m -strongly convex over $\mathbb{R}^n$. Each local cost function f_i is differentiable, m -strongly convex, and its gradient is locally Lipschitz on $\mathbb{R}^n$. The nonlinear coupling function $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ satisfies:*

1. $g(0) = 0$
2. $s^\top g(s) \geq 0 \quad \forall s \in \mathbb{R}^n$, with equality if and only if $s = 0$. (monotonicity)
3. $g(-s) = -g(s)$ (odd symmetry)
4. g is locally Lipschitz on $\mathbb{R}^n$

Under Assumption 2, the strong convexity of the global cost function guarantees that the optimization problem admits a unique solution (see (1)).

Assumption 3. *The communication graph $\mathcal{G}$ is undirected. The incidence matrix D of $\mathcal{G}$ satisfies $\mathbf{1}_N^\top D = 0$, which ensures the invariance of $\sum_i v^i(t)$ under the nonlinear coupling.*

Assumption 4. *The initial conditions of the internal states of the agents sum to 0, i.e., $\sum_{j=1}^N v(0) = 0$.*

Remark 1. The invariance of $\sum_{i=1}^N v^i(t) = 0$ is preserved under the nonlinear coupling g . This follows from the fundamental property $\mathbf{1}_N^\top D = 0$ of the incidence matrix, which holds independently of the choice of g . Hence Assumption 4 is compatible with the nonlinear modification.

Assumption 5. *The nonlinear coupling function g admits a potential function $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ such that:*

$$\nabla_s \Phi(s) = g(s), \quad \Phi(0) = 0, \quad \Phi(s) \geq 0 \quad \forall s \in \mathbb{R}^n.$$

Under Assumption 2, the optimization problem (3) can be reformulated in a distributed form as

$$\min_{x^i \in \mathbb{R}^n} f(x^i) = \min_{x^i \in \mathbb{R}^n} \sum_{i=1}^N f_i(x^i), \quad \text{subject to } (D^\top \otimes I_n)x = 0 \quad (4)$$

where x^i is the state of agent i , and $x = [x^1 \ \cdots \ x^N]^\top \in \mathbb{R}^{Nn}$ denotes the network state, and D is the incidence matrix of $\mathcal{G}$.

The constraint $(D^\top \otimes I_n)x = 0$ represents the consensus condition, since it encodes the pairwise disagreement across every edge of the graph. Specifically:

$$\begin{aligned} (D^\top \otimes I_n)x &= 0 \\ \iff x^i &= x^j \quad \forall (i, j) \in \mathcal{E} \\ \iff \exists \tau &\in \mathbb{R}^n \text{ such that } x = \mathbf{1}_N \otimes \tau, \end{aligned}$$

where the last equivalence holds since $\mathcal{G}$ is strongly connected. Note that this constraint is equivalent to the Laplacian-based condition $(L \otimes I_n)x = 0$, since $L = DD^\top$, we have

$$L \otimes I_n = (D \otimes I_n)(D^\top \otimes I_n).$$

Therefore,

$$x^\top (L \otimes I_n)x = x^\top (D \otimes I_n)(D^\top \otimes I_n)x = \|(D^\top \otimes I_n)x\|^2.$$

Since $L \otimes I_n$ is positive semi-definite,

$$\begin{aligned} (L \otimes I_n)x = 0 &\iff x^\top (L \otimes I_n)x = 0, \\ x^\top (L \otimes I_n)x = 0 &\iff \|(D^\top \otimes I_n)x\|^2 = 0, \\ \|(D^\top \otimes I_n)x\|^2 = 0 &\iff (D^\top \otimes I_n)x = 0. \end{aligned}$$

However, the incidence matrix formulation is more natural for the nonlinear coupling $g(D^\top x)$ considered in this work, since the nonlinearity acts directly on the edge disagreements $x^i - x^j$ rather than on the node states. This reformulation makes the problem suitable for distributed algorithm design and analysis.

3 Distributed Optimization over Undirected Graphs with Nonlinear Edge-Based Coupling

In this section, we present a fully distributed algorithm for solving the optimization problem in Equation (4) with a nonlinear edge-based coupling function g satisfying Assumption 2.

For each agent $i \in \{1, 2, \dots, N\}$, consider the dynamics

$$\dot{v}^i = \alpha\beta \sum_{j=1}^N a_{ij} \text{sign}(x^i - x^j), \quad (5a)$$

$$\dot{x}^i = -\alpha \nabla f_i(x^i) - \beta \sum_{j=1}^N a_{ij} \text{sign}(x^i - x^j) - v^i, \quad (5b)$$

where $v^i \in \mathbb{R}^n$ denotes the internal state of agent i , and $\alpha, \beta \in \mathbb{R}_{>0}$ are static control gains. The algorithm is distributed in the sense that each agent updates its state using only its own information and information received from its immediate neighbors.

Let $\mathcal{D} := D \otimes I_n$, then the overall network dynamics can be written compactly as

$$\dot{v} = \alpha\beta \mathcal{D} \text{sign}(\mathcal{D}^\top x), \quad (6a)$$

$$\dot{x} + \beta \mathcal{D} \text{sign}(\mathcal{D}^\top x) + v \in -\tilde{f}(x), \quad (6b)$$

where D is the incidence matrix of $\mathcal{G}$, $v = [v^1 \ \dots \ v^N]^\top \in \mathbb{R}^{Nn}$, and

$$\tilde{f}(x) = [\partial f_1(x^1) \ \dots \ \partial f_N(x^N)]^\top \in \mathbb{R}^{Nn},$$

$$\text{with } \partial f_i(x^i) \in \mathbb{R}^n \quad \forall i \in \{1, 2, \dots, N\}.$$

Here, g is applied elementwise to the vector $(D^\top \otimes I_n)x \in \mathbb{R}^{|\mathcal{E}|n}$, whose e -th block entry corresponds to the pairwise disagreement $x^i - x^j$ across edge $e = (i, j) \in \mathcal{E}$. In matrix form, the network dynamics are given by

$$\begin{pmatrix} \dot{v} \\ \dot{x} \end{pmatrix} = \begin{pmatrix} 0 & \alpha\beta \mathcal{D} \\ -I & -\beta \mathcal{D} \end{pmatrix} \begin{pmatrix} v \\ \text{sign}(\mathcal{D}^\top x) \end{pmatrix} - \alpha \begin{pmatrix} 0 \\ \nabla \tilde{f}(x) \end{pmatrix}.$$

Introducing the stacked state variable

$$z := \begin{pmatrix} v \\ x \end{pmatrix},$$

define

$$h(x, v) := \begin{pmatrix} \alpha\beta \mathcal{D} \text{sign}(\mathcal{D}^\top x) \\ -v - \beta \mathcal{D} \text{sign}(\mathcal{D}^\top x) \end{pmatrix}, \quad F(x) := \begin{pmatrix} 0 \\ \nabla \tilde{f}(x) \end{pmatrix}.$$

The network dynamics can then be written compactly as

$$\dot{z} = h(x, v) - \alpha F(x). \quad (7)$$

The vector field in (7) is discontinuous on the set

$$\left\{ z : (\mathcal{D}^\top x)_i = 0 \text{ for some } i \right\},$$

owing to the presence of the componentwise sign function. To obtain a well-defined description of the dynamics at these discontinuity points, the system is interpreted in the sense of a differential inclusion by replacing $\text{sign}(\cdot)$ with its set-valued extension $\text{Sgn}(\cdot)$, defined componentwise as

$$\text{Sgn}(y_i) = \begin{cases} \{1\}, & y_i > 0, \\ [-1, 1], & y_i = 0, \\ \{-1\}, & y_i < 0. \end{cases} \quad (8)$$

Accordingly, the single-valued mapping $h(x, v)$ is replaced by the set-valued mapping

$$H(x, v) := \begin{pmatrix} \alpha\beta \mathcal{D} \text{Sgn}(\mathcal{D}^\top x) \\ -v - \beta \mathcal{D} \text{Sgn}(\mathcal{D}^\top x) \end{pmatrix}. \quad (9)$$

The corresponding componentwise set-valued dynamics are therefore

$$\dot{v} \in \alpha\beta \mathcal{D} \text{Sgn}(\mathcal{D}^\top x), \quad (10)$$

$$\dot{x} \in -v - \beta \mathcal{D} \text{Sgn}(\mathcal{D}^\top x) - \alpha \nabla \tilde{f}(x). \quad (11)$$

Equivalently, defining the set-valued map

$$\mathcal{F}(z) := H(x, v) - \alpha F(x), \quad (12)$$

the complete network dynamics admit the differential-inclusion representation

$$\dot{z} \in \mathcal{F}(z). \quad (13)$$

Thus, the original discontinuous dynamics in (7) are represented by the set-valued dynamics (13), which explicitly account for the discontinuities induced by $\text{sign}(\mathcal{D}^\top x)$.

Remark 2. When $g(s) = s$, the compact form (6) reduces to the standard linear consensus-based algorithm, since $(D \otimes I_n)(D^\top \otimes I_n) = L \otimes I_n$.

3.1 Invariance of the Auxiliary Variable

A natural structural property of the compact formulation (6) is the invariance of the auxiliary variable v , namely $\sum_{i=1}^N v^i(t) = 0$ for all $t \geq 0$. Pre-multiplying (6a) by $(\mathbf{1}_N^\top \otimes I_n)$ gives

$$(\mathbf{1}_N^\top \otimes I_n) \dot{v} = \alpha \beta (\mathbf{1}_N^\top D \otimes I_n) g\left((D^\top \otimes I_n)x\right) = 0,$$

where the last equality follows from the fundamental property $\mathbf{1}_N^\top D = 0$ of the incidence matrix, which holds for any choice of g satisfying Assumption 2. Therefore, invoking Assumption 4,

$$\sum_{i=1}^N v^i(t) = \sum_{i=1}^N v^i(0) = 0, \quad \forall t \geq 0.$$

3.2 Equilibrium Points

Lemma 3 (Equilibrium Points). *Every equilibrium point of the system (6) is a consensus configuration at the unique global minimizer of (4), i.e.,*

$$\bar{x} = \mathbf{1}_N \otimes x^*, \quad x^* = \arg \min_x \sum_{i=1}^N f_i(x).$$

Proof. Setting $\dot{v} = 0$ and $\dot{x} = 0$ and denoting an equilibrium by $(\bar{x}, \bar{v})$. From (6a), $\dot{v} = 0$ requires

$$(D \otimes I_n) g\left((D^\top \otimes I_n)\bar{x}\right) = 0.$$

Since $g(0) = 0$ and g is monotone by Assumption 2, this implies

$$(D^\top \otimes I_n)\bar{x} = 0,$$

which by the strong connectivity of $\mathcal{G}$ gives

$$\bar{x} = \mathbf{1}_N \otimes \tilde{\theta}, \quad \tilde{\theta} \in \mathbb{R}^n.$$

From (6b), $\dot{x} = 0$ and the previous step give

$$\bar{v} = -\alpha \tilde{f}(\bar{x}),$$

or equivalently $\bar{v}^i = -\alpha \nabla f_i(\tilde{\theta})$ for all $i \in \{1, \dots, N\}$.

Summing over all agents and applying the invariance $\sum_{i=1}^N \bar{v}^i = 0$ established above gives

$$\sum_{i=1}^N \bar{v}^i = -\alpha \sum_{i=1}^N \nabla f_i(\tilde{\theta}) = 0,$$

which yields

$$\nabla \left(\sum_{i=1}^N f_i(\tilde{\theta}) \right) = 0.$$

By Assumption 2, the global cost function is m -strongly convex, so its minimizer is unique, giving $\tilde{\theta} = x^*$. Substituting back yields $\bar{x} = \mathbf{1}_N \otimes x^*$, which completes the proof. $\square$ $\square$